

Paper proposal: Particle and system dependence of correlations in STAR at RHIC

- PAs: Christine Nattrass, Jana Bielcikova, Betty Abelev
- Study of near-side Jet and Ridge components of h-V0 and V0-h correlations in Cu+Cu, Au+Au at $\sqrt{s_{NN}} = 200$ GeV
- Likely will come out after energy and system dependence paper
- May add more detailed multistrange track merging section
- 9 figures proposed
- Beginning of analysis note web page:
<http://rhig.physics.yale.edu/~nattrass/Research/web/index.html>
- Target journal: PRC
- Planned as rapid communication (5 pages) if possible

Reasoning behind separating this from the energy and system dependence paper

The paper on energy and system dependence has a clear, concise message that is distinct from the goal of this paper. The energy and system dependence paper posits that for h-h correlations a lot can be understood by looking at fragmentation alone (hence comparisons to Pythia,) that there is a Ridge at 62 GeV, and that the ratio of the Ridge/Jet, at least in the kinematic range we can study, is independence of collision energy and collision system. The message of this paper is that there is no system dependence for strangeness and no strangeness content dependence.

This paper involves strange particles, for which fragmentation is still poorly understood. Pythia comparisons would distract from the message in this paper and strangeness studies and details of identifying strange particles and correcting for track merging for the multistrange Xi would distract from the message in that paper.

Abstract

Paper Proposal: Particle and system dependence of correlations in STAR at RHIC

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Two-particle high- p_T triggered correlations involving strange particles in $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}} = 200\text{ GeV}$ in STAR at RHIC are presented. The N_{part} , $p_T^{associated}$, and $p_T^{trigger}$ dependence of the yield per trigger in the *Jet* and *Ridge* components of the near-side is investigated for K_S^0 , Λ , and Ξ trigger and K_S^0 and Λ associated particles. The baryon/meson ratio in the *Jet* is investigated. No dependence on the collision system is observed and no strong dependence on strangeness content is observed, in agreement with previous studies. These data strengthen the argument that the dominant production mechanism for the *Jet* is vacuum fragmentation, and constrain models which would predict a strong dependence on particle trigger type or strangeness content.

PACS numbers: 25.75.-q, 21.65.Qr, 24.85.+p, 25.75.Bh

Figure 1

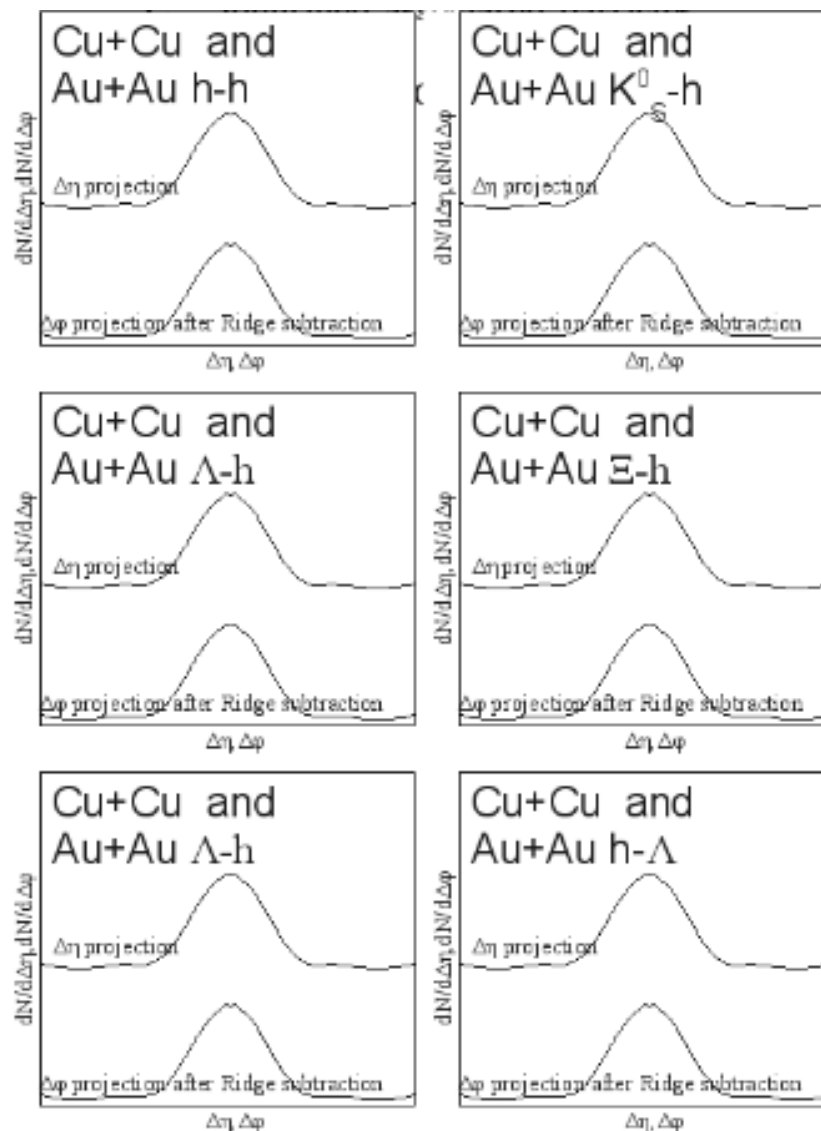


FIG. 1: To be filled in with real data - samples of Jet correlations from each particle type and system. $\Delta\eta$ projection before and after Ridge subtraction. Color online.

Figure 2

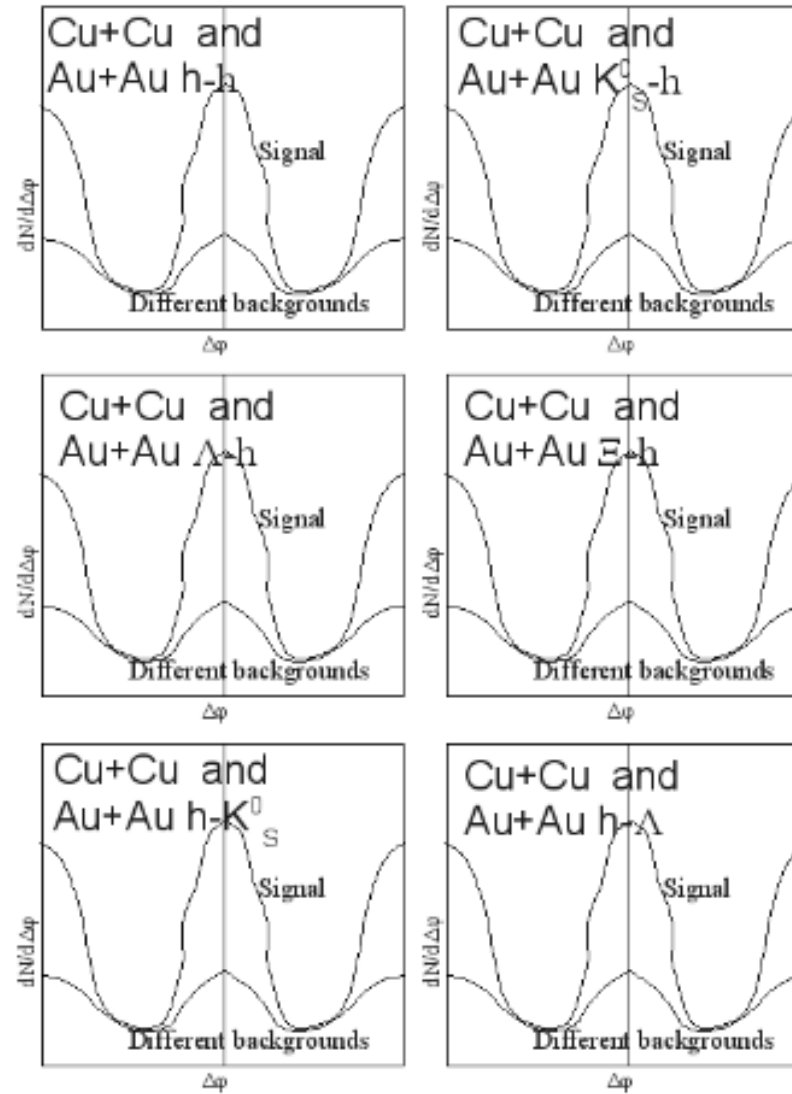


FIG. 2: To be filled in with real data - samples of Jet correlations from each particle type and system. $\Delta\phi$ projection inclusive and with Ridge subtracted off. Color online.

Figure 3

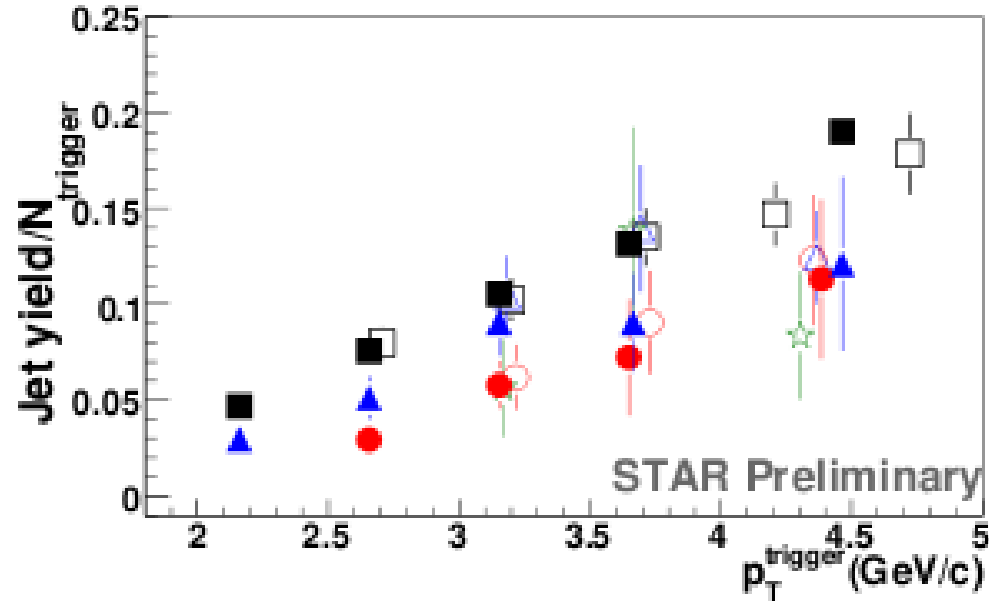


FIG. 3: Dependence of the *Jet* yield on p_T^{trigger} for identified trigger particles. No dependence on trigger particle type or strangeness content is observed. The *Cu + Cu* and *Au + Au* data are within error. Ξ points for *Au + Au* to be added to this plot, applying same method for *Jet/Ridge* separation as other particles with caveat that this might have a bias Color online.

Figure 4

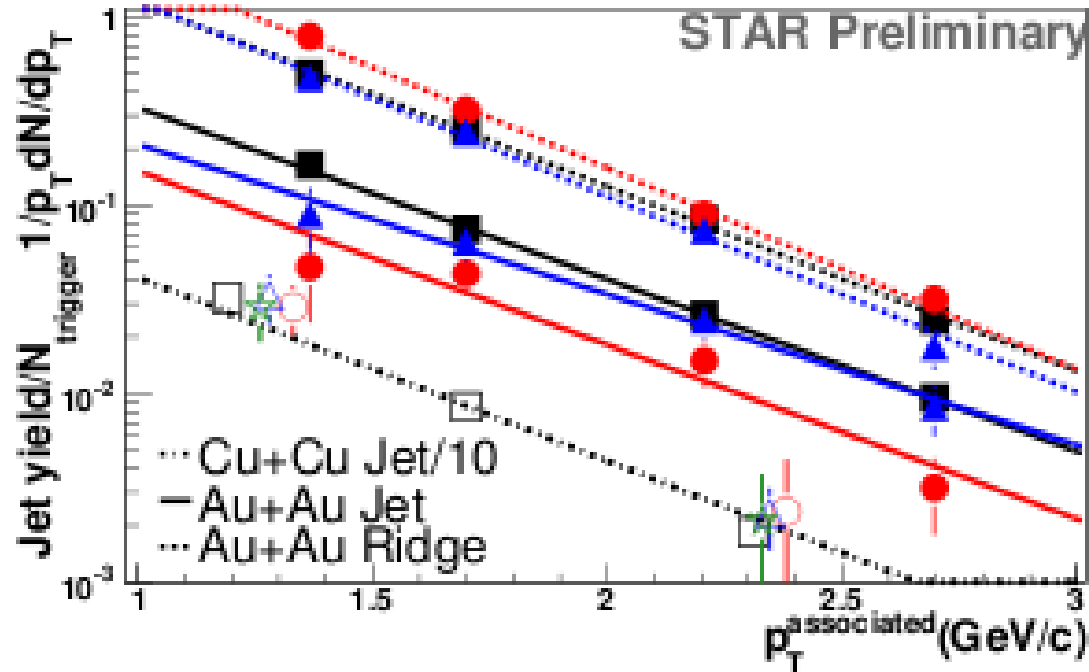


FIG. 4: Dependence of Jet yield on $p_T^{\text{associated}}$. Inverse slope parameters from fits in table I. No dependence on trigger particle type or strangeness content is observed.

The $Cu + Cu$ and $Au + Au$ data are within error. *If possible add Ξ yields.* Color online.

Figure 5

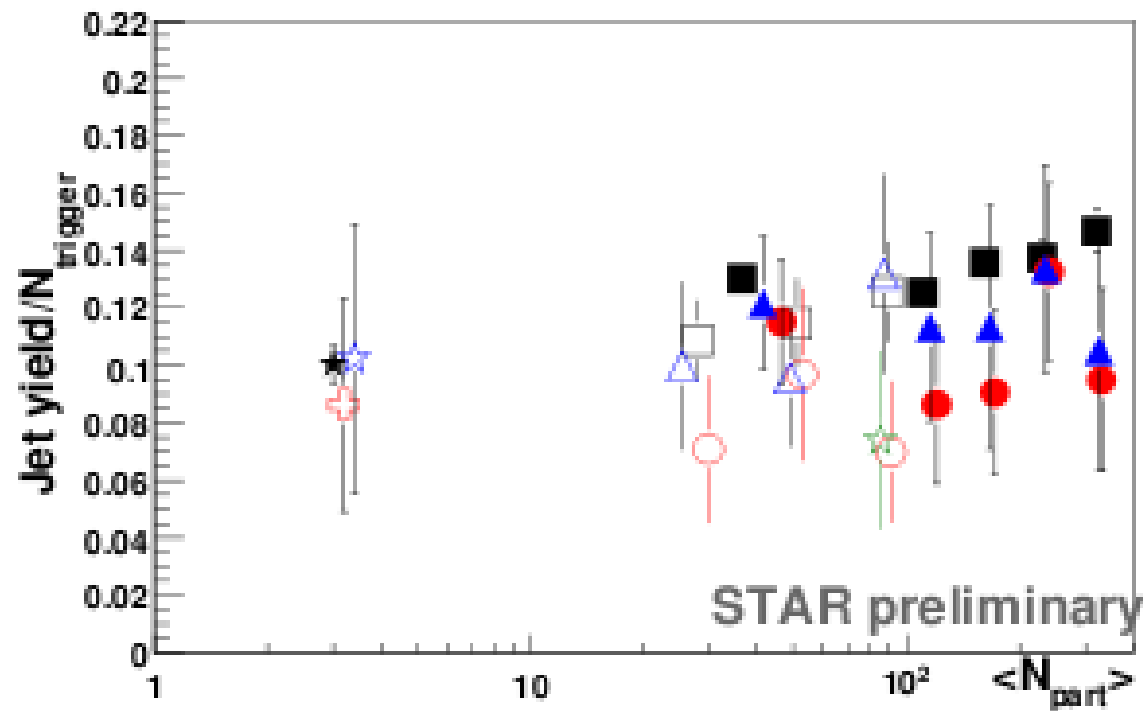


FIG. 5: Dependence of *Jet* yield on N_{part} . No or little, depending on how much track merging corrections move data points N_{part} dependence is observed. The $Cu + Cu$ and $Au + Au$ data are within error. Color online.

Figure 6

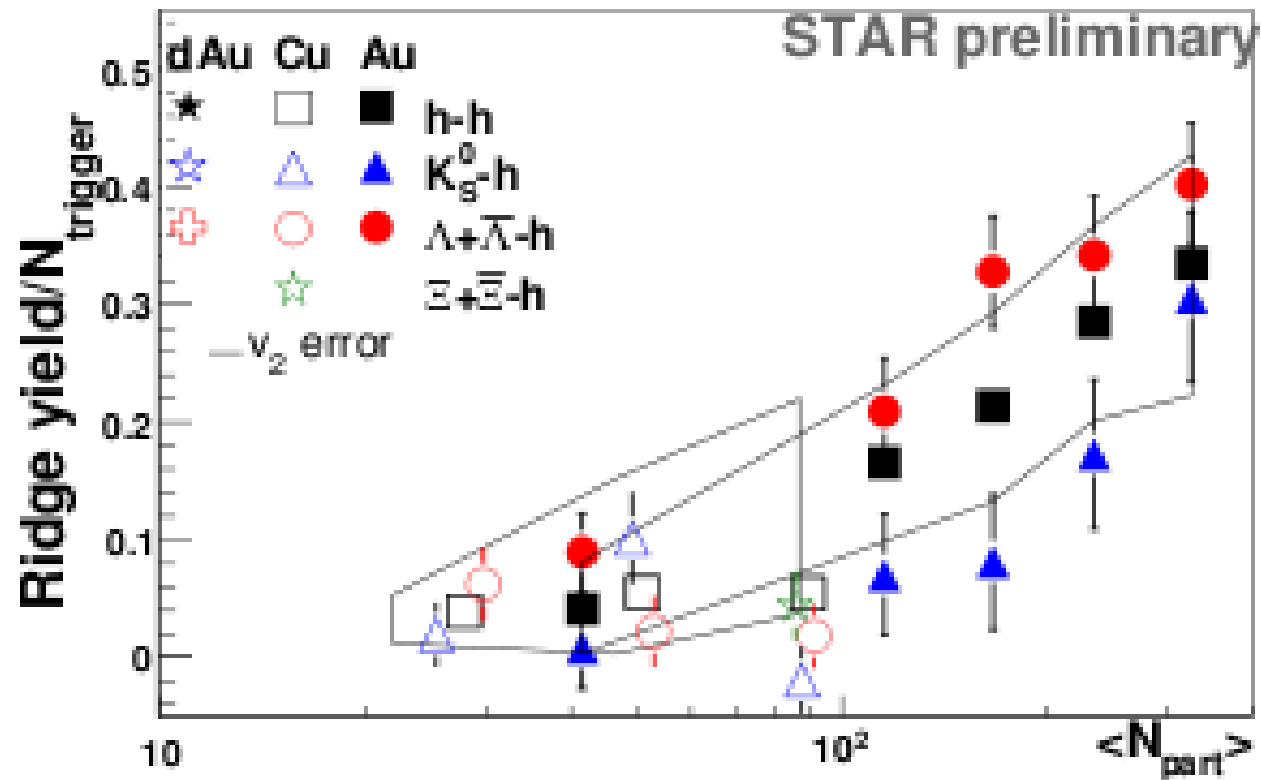


FIG. 6: Dependence of *Ridge* on N_{part} . The $Cu + Cu$ and $Au + Au$ data are within error. Color online.

Figure 7

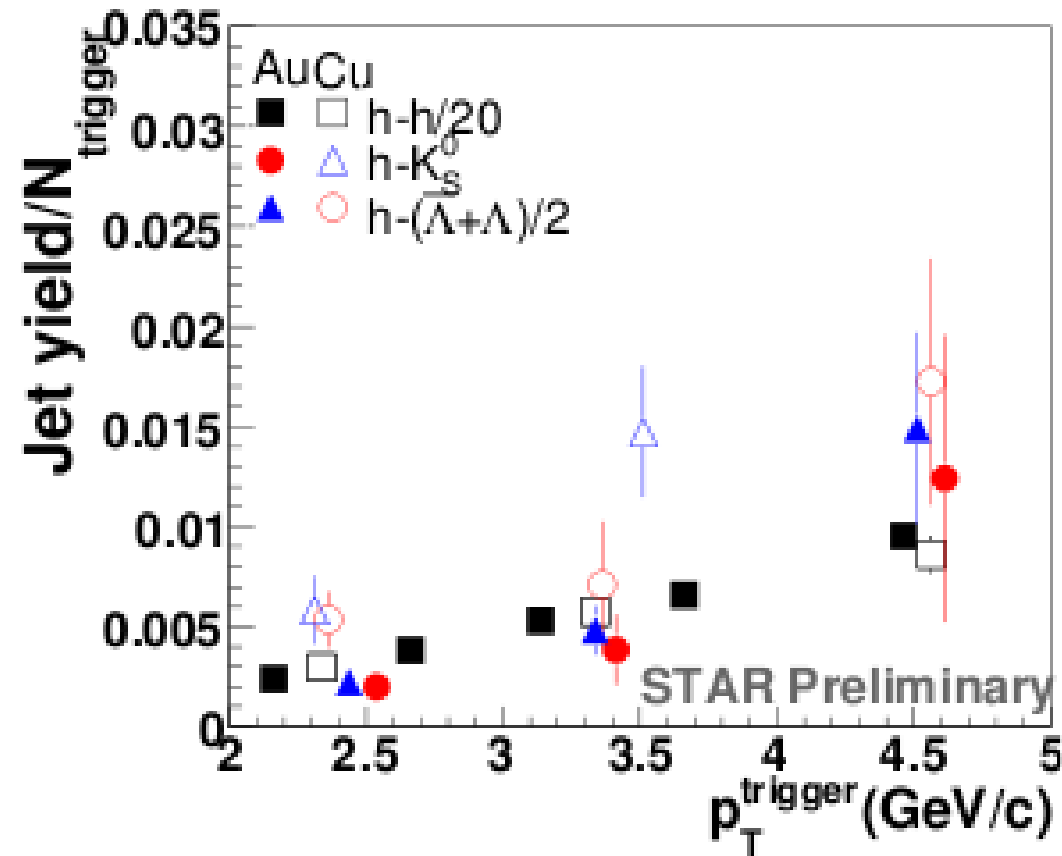


FIG. 7: Dependence of *Jet* yield on p_T^{trigger} . The $Cu + Cu$ and $Au + Au$ data are within error. Color online.

Figure 8

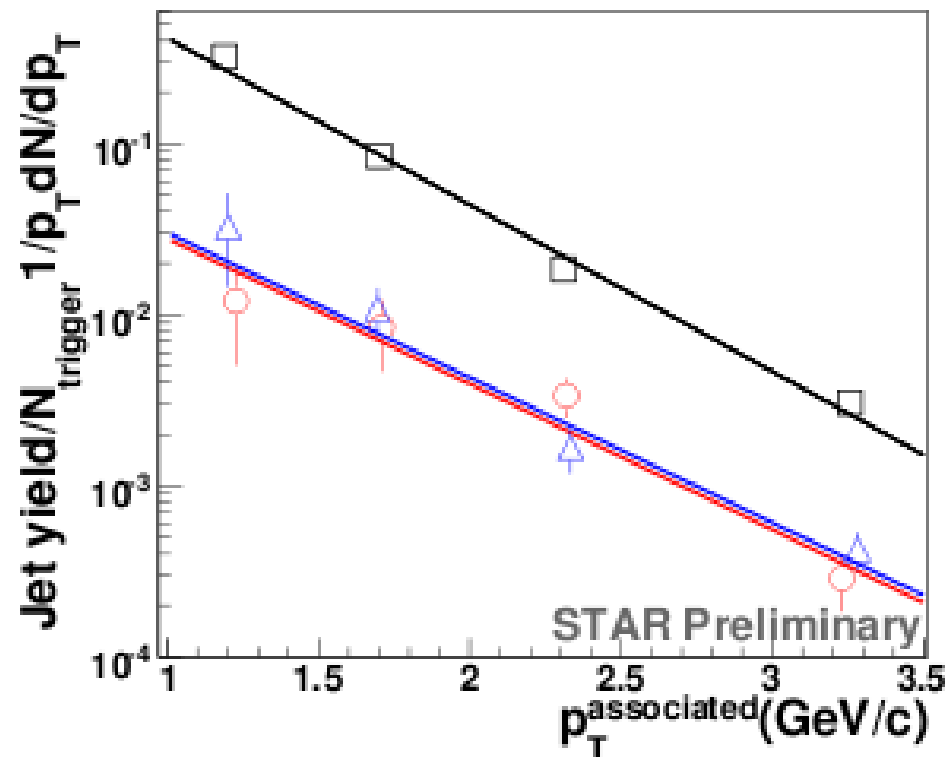


FIG. 8: Dependence of *Jet* yield on $p_T^{\text{associated}}$. Add *Au + Au* data? The baryon/meson ratio of 1 in the *Jet* persists across a wide $p_T^{\text{associated}}$ range, consistent with inclusive data from *p + p* add citation. Inverse slope parameters from fits in table I. Color online.

Figure 9

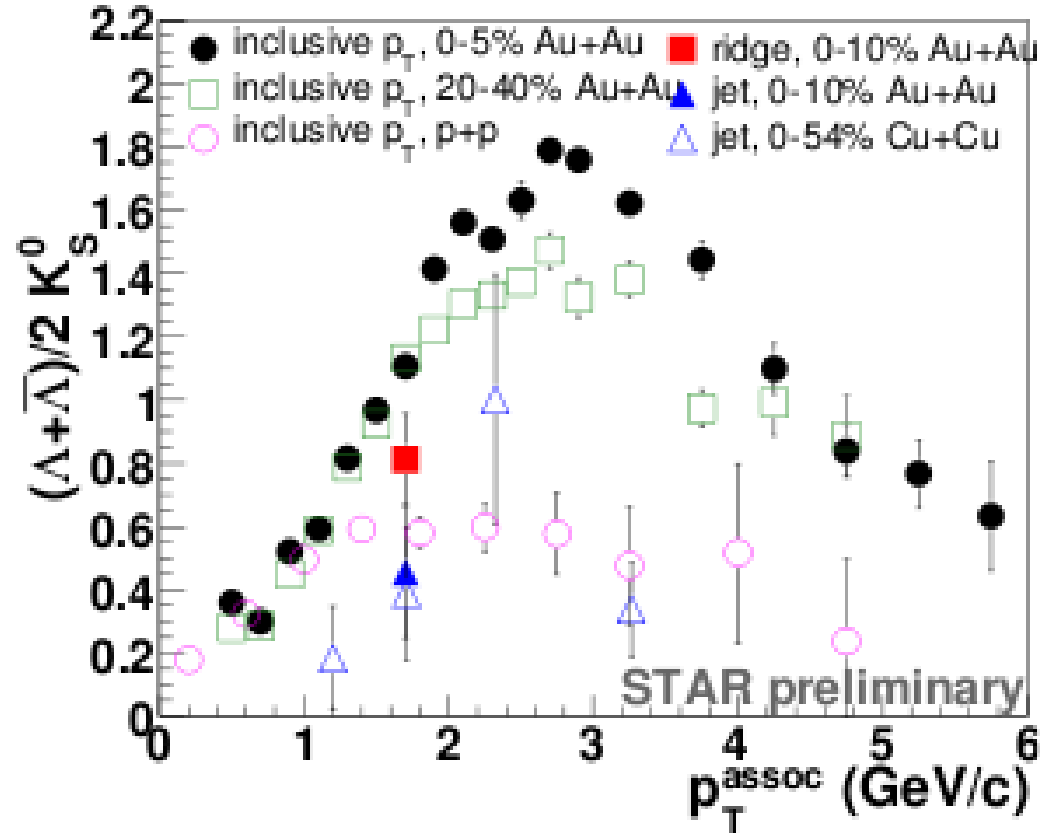


FIG. 9: Comparison of $\frac{\Lambda + \bar{\Lambda}}{2 K_S^0}$ in inclusive spectra and in *Jet* and *Ridge* region. Color online.

Conclusions

III. CONCLUSIONS

Neither the *Jet* nor the *Ridge* yields show dependence on trigger particle type or strangeness content. There is *Jet* component of the near-side of high- p_T correlations is fragmentation, although there is room within the errors for small contributions from other mechanisms. These data limit the production mechanisms feasible for the *Ridge* to those where there is no strong dependence on system or particle type.

To-do list

IV. TO-DO LIST

1. Implement track merging correction and run over data again. This is because the older version of my TTrees does not contain enough information to do the track merging correction, but thanks to Marek & Leon the method is developed and just needs to be implemented.
2. Contact theorists to ask them what they think their models would predict to flesh out the discussion section
3. After track merging corrections are done play with getting yield from $\Delta\eta$ instead, bin counting
4. Analysis note and plots of correlations which yield each data point